

Robust Digitization of Perioperative Surgical Flowsheets From Low and Middle Income Countries

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Abstract—In many low- and middle-income countries, health-care providers continue to rely heavily on paper-based medical records for data collection. This reliance stems from multiple barriers, including the high costs and infrastructure demands associated with implementing and maintaining electronic medical record systems and automated data capture technologies. Consequently, clinical data documented on paper remains largely inaccessible in digital form, in contrast to the availability of real-time digital data common in high-income settings. This lack of digitization presents a significant obstacle for clinicians, researchers, and quality improvement initiatives, impeding efforts to leverage data for improving patient care and outcomes. To address these challenges, the *ChartExtractor* program was created to automate the digitization of paper anesthesia charts. In this study, we introduce several enhancements to the system through the integration of advanced image processing and machine learning techniques. Unlike previous methods, this work combines thin plate splines and non-rigid point registration to enhance image alignment. Furthermore, we leverage multiple clustering techniques to accurately label the axes of intraoperative vital sign charts, making it easier to extract the corresponding values. Lastly, we used hyperparameter tuning to select smaller, more accurate object detection models, reducing inference time by 54% and deployment size by 183 MB. These advancements lay the groundwork for the *EQUAL Anaesthesia* mobile health platform, a low-cost solution designed to support digital data collection and clinical decision-making in resource-constrained healthcare settings.

Index Terms—Computer vision, Feature detection, Anesthesia, Mobile application

I. INTRODUCTION

The surgical mortality rates in low- and middle-income countries (LMICs) are double that of high-income countries (HICs), even though patients operated on in LMICs are typically at lower risk [1]. Preventing adverse surgical and anesthesia care outcomes relies on continual patient surveillance and care quality improvements. In an operation setting, data is collected recurrently, transcribing patient vitals, medication administration, and other surgical information. Driven by a digital transformation in healthcare, HICs have shifted from paper-based records to Electronic Medical Record systems (EMRs) for efficient data accessibility, real-time monitoring, and post-operative analysis. In contrast, many LMICs still rely on paper-based medical records to track perioperative data, which creates challenges for providers to later identify areas of improvement in care. [2]. The major resource dependencies of EMRs make them an unrealistic solution for digitizing LMIC data in the

foreseeable future, and manual digitization is time-consuming and unsustainable [3].

In this paper, we propose enhancements to the *ChartExtractor* computer vision program for digitizing surgical flowsheets in collaboration with the University of Virginia (UVA) Department of Anesthesiology [4]. The proposed work outlines novel and improved aspects of this software through intraoperative vital sign chart label identification, object detection model compression and improvement, thin plate spline registration, and coherent point drift registration.

II. RELATED WORK

Recently, several techniques have been developed to digitize paper perioperative sheets. Early efforts in this direction included the use of a wooden box to ensure image quality [5], detection of blood pressure symbols through the use of U-Nets [6], and detection of checkboxes with ORB template matching [7]. Although these techniques showed promising results, they often faced challenges related to operational use, including high processing time and insufficient image preprocessing.

More recent approaches have employed machine learning and advanced image processing techniques to overcome these challenges. For example, Annaparedy et al. deployed a YOLOv5 single-shot detector with a separate convolutional neural network (CNN) to classify detections [8]. In their pipeline, they used ORB template matching to extract relevant sections of the perioperative flowsheet and promote the homogeneity among inputs. This previous work contained a series of limitations that can prohibit the real-world deployment of the system. First, the YOLOv5 models consisted of high memory usage due to their size. Additionally, the classification models used specifically classify single digits despite the perioperative flowsheets containing multi-digit values. Finally, the image preprocessing steps taken were linear in fashion and failed to consider non-linear warping that may be necessary during operational use. This work builds upon this previous research work by improving the deep learning models, object classification, and image registration to allow real-world deployment.

III. DATA DESCRIPTION

Pre-existing work on this program resulted in the creation of a standardized anesthesia chart with the intent to move away from hospital site-specific flowsheets and reduce the amount

of handwritten words [4]. Documentation is separated into intraoperative and preoperative/postoperative sides. Electronic templates of these sheets have been created and printed for data creation. A team of resident and attending anesthesiologists populated flowsheets with biologically plausible but fabricated data and captured smartphone images of these charts to simulate user-submitted images. These images were then annotated with ground-truth labels.

A. The Pre and Postoperative Side

Information is collected predominantly through the use of checkboxes, replacing handwritten words and digits. Preoperative patient data includes the patient profile, medical history, and preoperative evaluation. Lower on the chart is the postoperative data section used to evaluate post-surgical recovery and an open section for additional notes.

B. The Intraoperative Side

The intraoperative side includes drug administration information as well as devices, tubes, and patient position. Freehand markings fill the chart's intraoperative side more in comparison to the pre/postoperative, which causes a greater focus on these more variable portions. The intraoperative side also contains the intraoperative vital sign chart, consisting of handwritten markings that encode heart rate and blood pressure captured at regular time intervals during an operation, seen in Fig. 1.

IV. METHODOLOGY

The following subsection outlines the methodology employed in our proposed approach. Organized by task, each subsection provides a detailed delineation of the techniques implemented for each specific optimization or addition.

A. Clustering Digits for Blood Pressure and Heart Rate Graph Interpretation

Computer vision models locate objects within an image and classify them based on objects they have been trained on. A computer vision model integrated in *ChartExtractor* was trained to detect *single* digits. However, the intraoperative vital sign chart contains multi-digit labels on the graph's axes that need to be identified and classified properly to impute meaning to the chart markings. Single digit classification fails to provide this semantic meaning (such as detecting '1', '2', and '0' individually instead of '120mmHg/bpm'). Determining the location of the axis labels on real-world images allows the extraction of highly accurate blood pressure and heart rate measurements at exact timestamps. A process was developed to combine detected digits and create axes landmarks, allowing for the extraction of data from the blood pressure and heart rate graph.

1) *Detection Selection*: Ground truth labels for the intraoperative vital sign chart axes were manually generated on 19 fabricated flowsheets, for both single digits (for example, '1', '2', and '0') as well as appropriately clustered numbers and units ('120mmHg/bpm'). Object detections were stored as bounding boxes (BBs), a rectangle framing an object or detection to mark the location and classification of the object

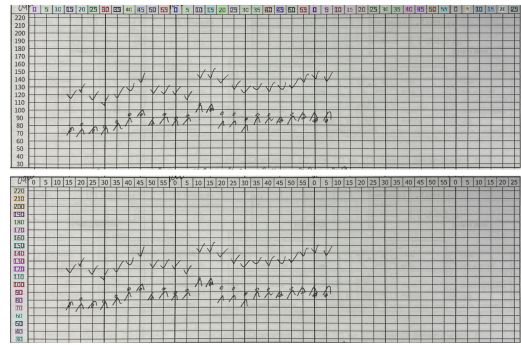


Fig. 1: Final clusters for each axis on a sample flowsheet.

identified. Each axes BBs were isolated and separated. This was done via maximum density measurements used to identify the graph's top-left corner coordinates and, therefore, its axes. Axis BB lists were used as input for different clustering methods to group single digits into their full numeric value.

2) *Clustering Methods Utilized*: Three clustering methods were developed and tested to group single digits into meaningful labels. Each is outlined below.

- *K-Means Clustering*: K-Means is implemented by supplying a list of detected BBs and a range of plausible total clusters. The number of labels on each axis is known, but to manage the possibility of missed or erroneous detections, we fit a range of possible total clusters and selected the one that maximized the silhouette score by a certain threshold.
- *Agglomerative Clustering*: Agglomerative clustering operates similarly to K-Means, as a predetermined number of clusters can be provided to the algorithm. It takes the same inputs as K-Means and also uses a silhouette score to manage missed or erroneous detections.
- *Density-Based Spatial Clustering of Applications with Noise (DBSCAN) Clustering*: DBSCAN differs from K-Means and Agglomerative clustering methods; the number of expected clusters is not used in this algorithm. Rather, the hyper-parameters include the minimum number of samples to group and the maximum distance for two samples to be considered in each other's "neighborhood".

3) *Assigning Meaning to Clusters*: Two lists, organized by axis, of grouped individual digits were the output from the clustering algorithms. Due to this axis split, the clusters were automatically understood to represent either minutes or heart rate/mmHg values. New landmark BBs were drawn around the full axes numbers, and labels were based on left-to-right single-digit coordinate locations within each cluster. Examples of final bounding boxes on the blood pressure graph are shown in Fig. 1.

B. Retraining Object Detection Models

Previously, Ultralytics YOLO v11 object detection models were trained to localize key elements within hand-written flowsheets. However, the collective model size (220 MB) led to increased processing time in the data extraction pipeline

and changes in the project architecture to manage this increased storage and computational burden. We sought to retrain these models using more compact and computationally efficient architectures to enhance efficiency, reduce server costs, and even facilitate the usage of these models within the *EQUAL Anaesthesia* phone application.

1) *Retrained Models and Their Applications*: The following 7 models were retrained and underwent hyperparameter tuning:

- **Systolic and diastolic blood pressure marker**: Two separate pose detection models designed to accurately localize the systolic and diastolic blood pressure markers on the blood pressure diagram and identify the reference point necessary for the precise determination of the mmHg.
- **Heart rate marker**: A pose detection model for localizing the heart rate marker. This detection pinpoints the exact point for heart rate measurement at a timestamp.
- **Checkbox**: A classification model that locates checkboxes on the surgical flow-sheet and determines whether they are marked or not.
- **Digit recognition**: An object detection model capable of identifying and classifying both handwritten and printed single-digit numeric values across the document.
- **Landmark detection models**: Two separate models were trained to determine key landmarks across the flowsheet to facilitate spatial alignment and registration of the intraoperative and pre/postoperative sides.

2) *Training and Optimization Protocol*: The aforementioned ground-truth labeled chart images were split into train, validation, and test sets as outlined in Table I.

Using these splits, all models were retrained as YOLOv11 nano models [9]. Hyperparameter optimization was performed iteratively to enhance model performance. The training regimen included 200 epochs with a batch size of 64 images for training and 25 optimization iterations, each consisting of 50 training epochs for hyperparameter tuning. The final model was selected based on mean average precision (mAP 50-95), with the highest-performing model retained for deployment.

3) *Model Assessment*: Models were evaluated based on time, size, and accuracy improvement. Time improvement was evaluated on the same machine using Python's cProfile package. Size improvement was based on storage size in MB.

Model performance was tested by calculating accuracy, precision, recall, F1, and mean absolute error in mmHg or heart rate for systolic, diastolic, and heart rate detection across 6 test sheets the models are not exposed to in the training process.

TABLE I: Number of sheets allocated for train, validation, and test sets when tuning a respective object detection model.

Object Detection Model	Train	Validation	Test
Digits	23	5	6
Systolic Markers	33	5	7
Diastolic Markers	33	5	7
Heart Rate Markers	33	5	7
Checkboxes	5	2	3
Intraoperative Landmarks	13	3	4
Pre/Postoperative Landmarks	12	3	4

Checkbox detection is also evaluated on the same 6 test sheets with a similar metric aside for mean absolute error.

Digit and landmark detection models' accuracy was evaluated via the Ultralytics YOLO package's model evaluation functionality of the test set provided for training and tuning [9].

C. Improving Detection Classification with Non-Linear Registration

User-submitted sheets deviated visually from the scanned standardized flowsheet, commonly with an angled perspective, shadowed image, or uncropped photo. at times causing the model to misclassify detected objects. Solving this problem was a necessity for the *ChartExtractor* to perform properly so that the data stored in the digital database is truly what its label claims. The fault tolerance and robustness of the program were increased by detection verification using non-linear registration methods.

1) *Point Registration and Coherent Point Drift*: Point registration is a method that applies spatial transformations to a collection of input (source) points to match them to a set of ground-truth (destination) points. Landmark locations from a source anesthesia flowsheet compared to the ideal destination sheet are seen in Fig. 2 and their matching was executed using the coherent point drift (CPD) algorithm.

Using CPD, the destination points are stationary while the source point cloud is treated as a collection of Gaussian mixture model (GMM) centroids, transformed using an Expectation Maximization (EM) framework. CPD aims to determine the correspondence between the two point clouds and represents this as the posterior probability of a (source) GMM centroid given a destination data point [10]. CPD is robust through its emphasis on the coherent (unified) movement of one point cloud to match the other, especially necessary for our highly dense data sheets.

2) *Registration Implementation with Pycpd*: Twenty annotated synthetic patient sheets provided twenty sheets of object landmark source points. Positional coordinates from the detected objects were first normalized and then registered using the Python package pycpd, a purely numpy implementation of the CPD algorithm [11]. Of the three types of CPD registration, we passed the source and destination points to be aligned using a stacked deformable (non-linear) and affine registration. The new set of registered points was used with the destination points to

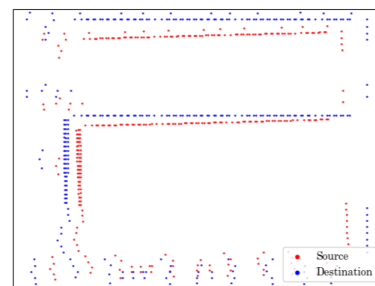


Fig. 2: Source and destination point clouds formed by detected bounding boxes on sheet images.

form a distance matrix, with each cell as the Euclidean distance of the intersection's corresponding row (registered point) and column (destination point). The minimum distance of each row was identified, and the point classifications associated with that cell were paired.

Hyperparameter optimization was completed using Optuna [12], an automated tuning framework. The parameters responsible for the trade-off between the goodness of maximum likelihood fit and regularization, α as well as the width of the Gaussian kernel, β we tuned.

D. Augmenting Registration with Thin-Plate Splines (TPS)

Thin plate splines (TPS) is a spline-based technique allowing us to perform non-linear point registration. TPS uses a set of source points and destination points and calculates the transformation needed to align the source and destination points. Previously, there was a large variation in the location and shape of objects on the flowsheets when the sheet was folded or crumbled. The linear registration technique that was previously implemented was not capable of addressing these deformations adequately.

In our work, TPS was utilized to normalize folds and bends in chart images and closely align these user-captured images with an ideal electronic template of the flowsheet. To perform TPS, we map a unique list of source BB detections determined by our landmark detection models to destination BBs on the digitized template. We created a framework allowing for the optional downscaling of image size with the intent to decrease RAM utilization. Using these lists of source and destination points, we estimate the TPS transformation with scikit-learn's radial basis function kernel (RBF) and store this transformation in a grid. This grid of transformations is then expanded back to the original image size. Finally, we apply this non-linear transformation to our image, allowing the source points to align with the destination points.

V. RESULTS

A. Label Clustering

Clustering was evaluated by comparing the ground truth axes labels to constructed labels. To be compared, clusters from the ground truth and detected sets had to have the same label. If there was a match, then the mAP 50-95 was computed between the bounding boxes of the two clusters. The methods were evaluated first without any erroneous detections and then with a combined 2.5% removal and 2.5% addition of erroneous digit detections. Erroneous digit detections were to simulate

TABLE II: Comparison of mAP 50-95 with and without the introduction of erroneous bounding boxes for the time and mmHg/bpm axes of the intraoperative vital sign graph.

Method	W/o Errors		W/ Errors	
	Time mAP	Number mAP	Time mAP	Number mAP
K-Means	1.0	1.0	0.77	0.99
DBSCAN	1.0	1.0	0.81	0.99
Agglomerative	1.0	1.0	0.79	0.99

TABLE III: Size improvement with retrained object detection models.

Detection Model	Prior Size (MB)	New Size (MB)	Size Reduction (MB)
Digit detection	40	5	35
Checkbox detection	19	5	14
Systolic detection	41	6	35
Diastolic detection	41	6	35
Heart Rate detection	41	5	36
Intraoperative landmark detection	19	5	14
Pre/Postoperative landmark detection	19	5	14
Total	220	37	183

TABLE IV: Speed improvement with retrained object detection models without GPU hardware acceleration. All experiments were conducted on a personal laptop (HP Spectre x360) equipped with an Intel Core i7-1165G7 CPU, 16 GB of RAM, and integrated Intel Iris Xe Graphics, running Windows 11 (64-bit).

Detection Model	Prior Speed (secs)	New Speed (secs)	Speed Reduction (secs)
Digit detection	18	8	10
Checkbox detection	23	8	15
Systolic detection	29	15	14
Diastolic detection	29	15	14
Heart Rate detection	14	4	10
Intraoperative landmark detection	8	5	3
Pre/Postoperative landmark detection	10	5	5
Total	131	60	71

worst-case real-world errors based on the precision of our digit detection model. The results of our clustering methods are shown in Table II.

B. Object Detection Models

We were able to successfully train models that were able to significantly decrease model size, speed up processing, and, in many cases, improve accuracy. Table III summarizes the significant reduction in model size. Table IV shows a breakdown of speed improvements to the detection portion of the *ChartExtractor* pipeline. Overall, there was a 54% speed-up.

Despite the reduction in size and object detection time, performance improvements were observed in systolic, diastolic, heart rate, and digit detection, seen in Table V. Additionally, Table VI shows the high quality of performance for landmark detection models.

C. Non-Linear Registration

Evaluations of the registration were based separately on registered and destination paired point distance (mean squared error) or classification (verified if pairing classification matched). The hyperparameters responsible for adjusting deformable transformations were investigated. Fig. 3 displays an example sheet of registered points compared to the original input source and

TABLE V: Improvement of detection models after retraining and fine-tuning.

Detection Model	Accuracy		Precision		Recall		F1		Mean Error*	
	Old	New	Old	New	Old	New	Old	New	Old	New
Systolic	0.95	0.98	1.0	1.0	0.95	0.98	0.98	0.99	0.82	0.60
Diastolic	0.91	0.95	1.0	1.0	0.91	0.95	0.96	0.98	2.19	1.88
Heart Rate	0.85	0.94	1.0	1.0	0.85	0.94	0.92	0.97	2.49	2.03
Checkbox	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	n/a	n/a

TABLE VI: Performance of digit and landmark detection models.

Detection Model	MAP 50-95		Precision		Recall	
	Old	New	Old	New	Old	New
Digit	0.83	0.84	0.94	0.94	0.79	0.79
Intraoperative landmark	n/a	0.97	n/a	0.97	n/a	0.96
Pre/Postoperative landmark	n/a	0.99	n/a	0.96	n/a	0.99

destination clouds. The Optuna search space was set to a range of (0.0001, 10) for α and (1, 50) for β , with 100 trials per sheet and 20 sheets. Fig. 4 shows the optimal registration hyperparameters for each source sheet.

D. Thin-Plate Splines

No objective mechanism for scoring TPS output was developed at the time of this publication. Tables of computation improvements with image size reduction and visual results on a folded sheet are included in Table VII and Fig. 5.

Memory increment is the increase in RAM usage when performing TPS. Runtime is calculated on the same CPU machine. Both are calculated using Python's built-in profiler.

VI. DISCUSSION

A. Overview of Contributions and Improvements

This project presents improvements and additions to the development of a robust pipeline for the extraction of anesthesia flowsheet data from smartphone images. Four significant improvements were made in three core areas: intraoperative vital sign chart interpretation, object detection model optimization, and non-linear image registration. These enhancements work in tandem to enable the accurate, scalable, and efficient digitization of clinical flowsheets.

First, we introduced a novel method for grouping individual digit detections for label axes into meaningful numeric labels and bounding boxes. By developing implementations of

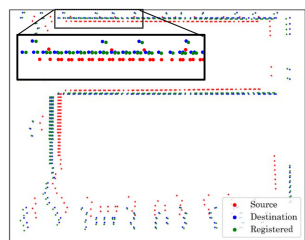


Fig. 3: Source, destination, and registered points.

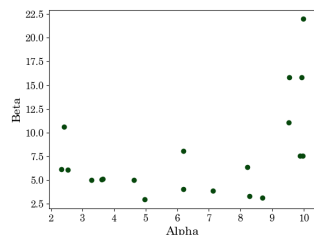
Fig. 4: Optimal α and β hyperparameters identified by optuna.

TABLE VII: Performance Improvement with Image Size Reduction for TPS

TPS Technique	Memory Usage Increase (RAM)	Runtime
No Image Size Reduction	753 MB	3.6 seconds
0.5x Image Size Reduction	187 MB	1.1 seconds

multiple clustering algorithms—including KMeans, DBSCAN, and Agglomerative Clustering—we were able to reliably group digits into numerical values while accounting for missed or spurious detections. This enabled the accurate assignment of time, heart rate (bpm), and blood pressure (mmHg) values to the detected handwritten symbols on the flowsheet.

Second, we retrained our object detection models using the lightweight YOLOv11 nano architecture, achieving a reduction in total model sizes from 220 MB to 37 MB. This drastically reduced processing time—a 54% speed-up overall—without compromising, and in some cases improving, detection accuracy. The reduced model footprint paves the way for future deployment on resource-limited hardware, such as within the *EQUAL Anaesthesia* phone application.

Third, landmark classification verification was investigated using non-linear registration of user-submitted flowsheet images matched to a standardized template. The Optuna hyperparameter tuning identified the CPD method's solution space to be non-smooth, with Fig. 4 highlighting the lack of convergence in either hyperparameter. Classification matching tests also verified a lack of convergence with an average of 22 misses across all registered sheets. In its current state, this registration method is not generalizable across input sheets but provides a robust framework for further research.

Lastly, to address non-linear distortions of chart images, we implemented TPS registration to augment the alignment achieved by rigid registration. To optimize for speed and memory usage, the transformation was computed on a 0.5x down-sampled version of the image, then this transformation matrix was expanded and applied at full resolution. This reduced memory usage from 753MB to 187MB and decreased runtime by 69%. More investigation is necessary to reach a consensus on the utilization of this technique in the flow of the *ChartExtractor*

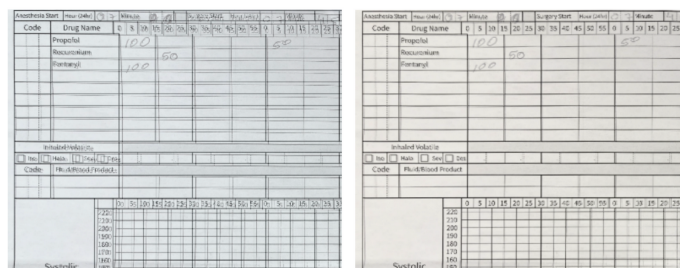


Fig. 5: Overlay of the electronic template sheet over the smartphone captured and linearly registered image before (left) and after (right) TPS with 0.5x downscaling. This figure is focused on the top left portion of the intraoperative side of the sheet for clarity.

software.

Together, these contributions represent a significant leap in the usability and reliability of extraction of chart data from smartphone-captured images. This has led to the groundwork for the development of a usable pipeline that can be integrated within the previously existing *ChartExtractor* framework. By combining computer vision techniques with registration and clustering methods, we enable structured, interpretable outputs from highly variable input images—bringing us closer to fully and reliably automating clinical data extraction.

B. Areas for Future Work

Despite the improvements and new contributions outlined above, a multitude of opportunities for improvement present themselves in each domain.

1) *Digit Clustering for Labels*: While our work in this domain sets a flexible basis for further development, plenty of work is necessary to ensure the robustness of the solution we present. Ideally, future work must use a more intelligent approach to assigning labels to clusters. Our current approach performs relatively poorly with erroneous detections due to the lack of a dynamic method for correcting incorrect digit value detections. If an inappropriately detected "7" is grouped into appropriately detected "1", "2", and "0" of the 120mmHg/bpm label on the Y-axis, we currently have no mechanism for correcting the label to 120mmHg/bpm by taking in the context of the value being located between 110 and 130 labels. This resulted in a drop in mAP 50-95 when erroneous bounding boxes were introduced.

Currently, values for what is determined as a "significant" increase in silhouette score and K-means, DBSCAN, and the various agglomerative clustering hyperparameters were manually chosen and optimized. These values can be tuned using a tuning framework such as Optuna on a larger set of data to ensure these values are optimized to improve performance across most cases.

2) *Consolidating and Improving Detection Models*: There may be increased room for improvement with detection model performance and computational burden by combining similar object detection models. For example, perhaps we can train a single model to detect all systolic and diastolic detections instead of using 2 separate models.

3) *Thin-Plate Splines for Augmented Image Registration*: TPS has significant potential to normalize images of distorted, folded, or crumpled papers to a near flat representation of the image. However, we currently need to develop a methodology for the objective assessment of the transformation that TPS performs. Then, using that, we can tune the parameters and the magnitude of image size reduction to improve performance. Once that is done we can conclusively determine if this methodology is practical in our final infrastructure.

4) *Non-linear Registration*: Further tuning is possible and necessary for the current non-linear registration implementation.

Considerations could be made to look outside the α and β hyperparameters to tune the method. Possible avenues include the noise parameter (w) of the EM registration class within the pycpd package, and normalization of different ranges for the source and destination points. Additionally, other point cloud registration methods could be studied.

VII. CONCLUSION

The novel additions and improvements made to the *ChartExtractor* software give significant momentum for future study. Our efforts increase the robustness and efficiency of the work through landmark label clustering, model consolidation, and image registration. This work helps complete the groundwork for a pipeline that can be integrated into the *EQUAL Anaesthesia* phone application for data extraction from paper charts.

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